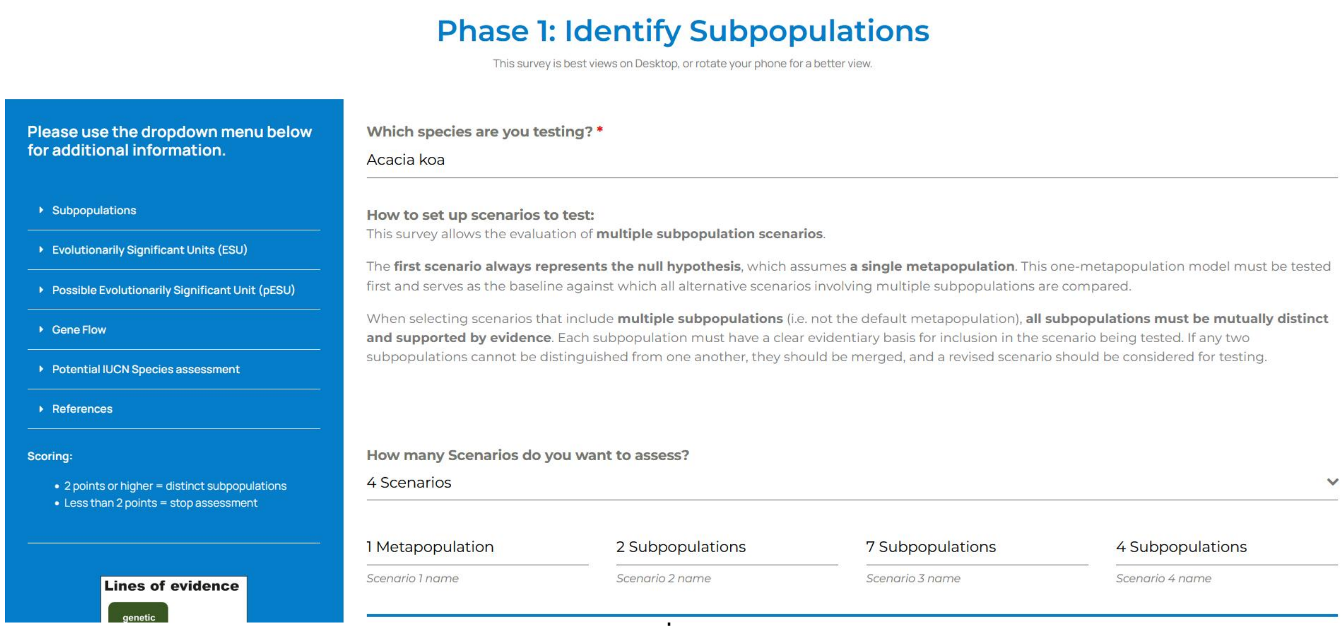
Detailed testing of the Hawaiian Koa (*Acacia koa*)

The Hawaiian koa is an endemic tree species of high ecological (e.g., provides key habitat for many threatened species and watershed recharge areas), economical (e.g., highly regarded hardwood timber) and cultural importance (e.g., timber is used to build traditional Hawaiian outrigger canoes for fishing, racing, and voyaging) (Baker et al., 2009; Mitchell et al., 2005). It is distributed on the four main Hawaiian Islands (the Island of Hawaiʻi, Maui, Oʻahu and Kauaʻi). Hawaiian koa grows in various habitats ranging from wet to drier areas in different elevations and therefore also expresses high variation in phenotypic traits (Baker et al., 2009). It is currently listed under the threat category ‘least concern’ in the IUCN Red list; however, the last assessment of koa was in 2010 (Contu, 2010). In 2015 an action plan for Hawaiian koa was developed to discuss the actions needed to achieve healthy koa forests that accommodate people and economic needs and promote forest conservation (Inman-Narahari, 2015).

Please refer to a detailed description of evidence used for the Hawaiian koa to DOI: 10.1093/biosci/biag042.

1. Fill in species and scenario information.
   We tested four different candidate scenarios: (1) a single metapopulation (null hypothesis), (2) two subpopulations based on genetic data (microsatellite markers; Fredua-Agyeman et al., 2008), (3) seven potential subpopulations, based on genomic data (genotyping-by-sequencing GBS; Gugger et al., 2018) and (4) four subpopulation based the geographic distribution on the four main Hawaiian oceanic islands.



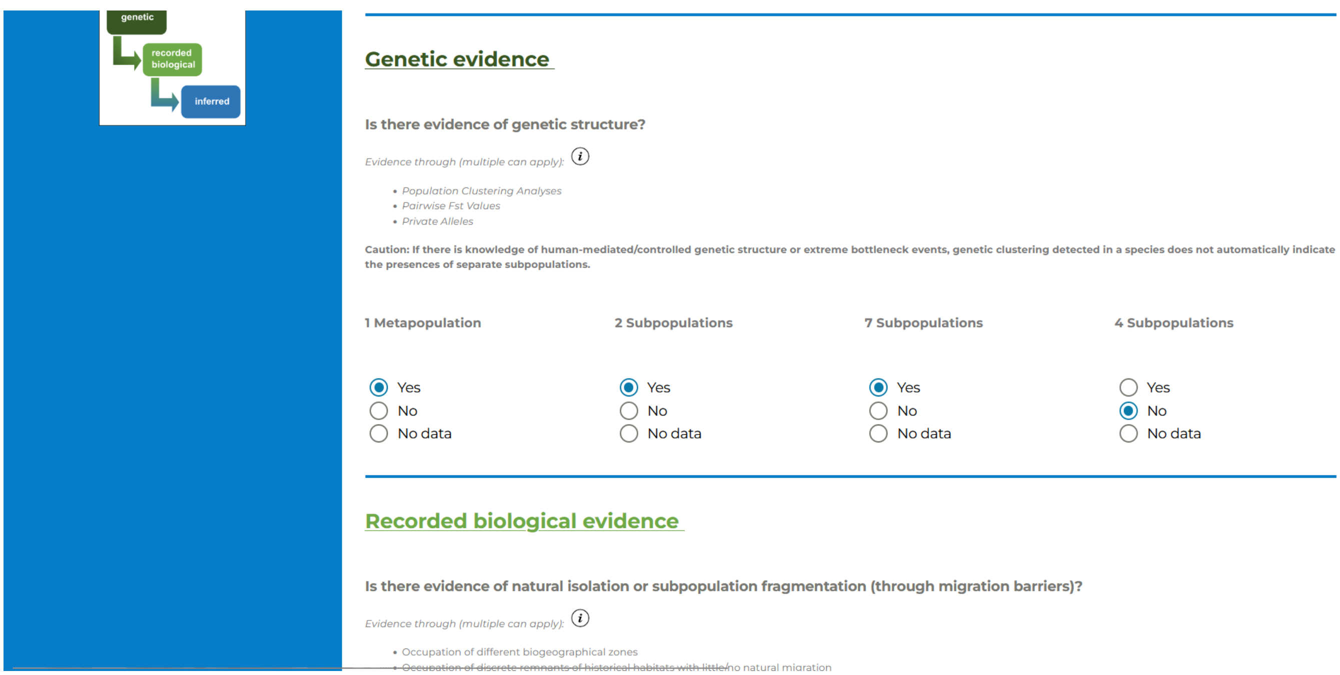
2. Answer the questions for each scenario.
    a. Genetic evidence

       In **Scenario (1)** (null hypothesis), depicting a single metapopulation. For this scenario scoring is reserved to keep the questions identical across all scenarios. The existing evidence did not indicate the presence of a single metapopulation of the Hawaiian koa. Therefore, a score of 0/2 for genetic structure was assigned.

       In **Scenario (2),** a genetic study identified two subpopulations (Kauaʻi vs. Island of Hawaiʻi, Maui, and Oʻahu) based on clustering analyses and pairwise FST values (Fredua-Agyeman et al., 2008). This resulted in a score of 2/2 for genetic structure.

       In **Scenario (3),** a genomic study identified seven subpopulations (one each on Kauaʻi, Maui, and Oʻahu, and four on the Island of Hawaiʻi) (Gugger et al., 2018). Despite some shared clusters across islands, these were attributed to recent human-mediated dispersal and limited gene flow was detected between the four oceanic islands of Hawai'i. This also resulted in a score of 2/2 for genetic structure.

       In **Scenario (4),** although four subpopulations could be inferred based on natural distribution, the genetic/genomic studies did not directly support this configuration. Therefore, a score of 0/2 for genetic structure was assigned.

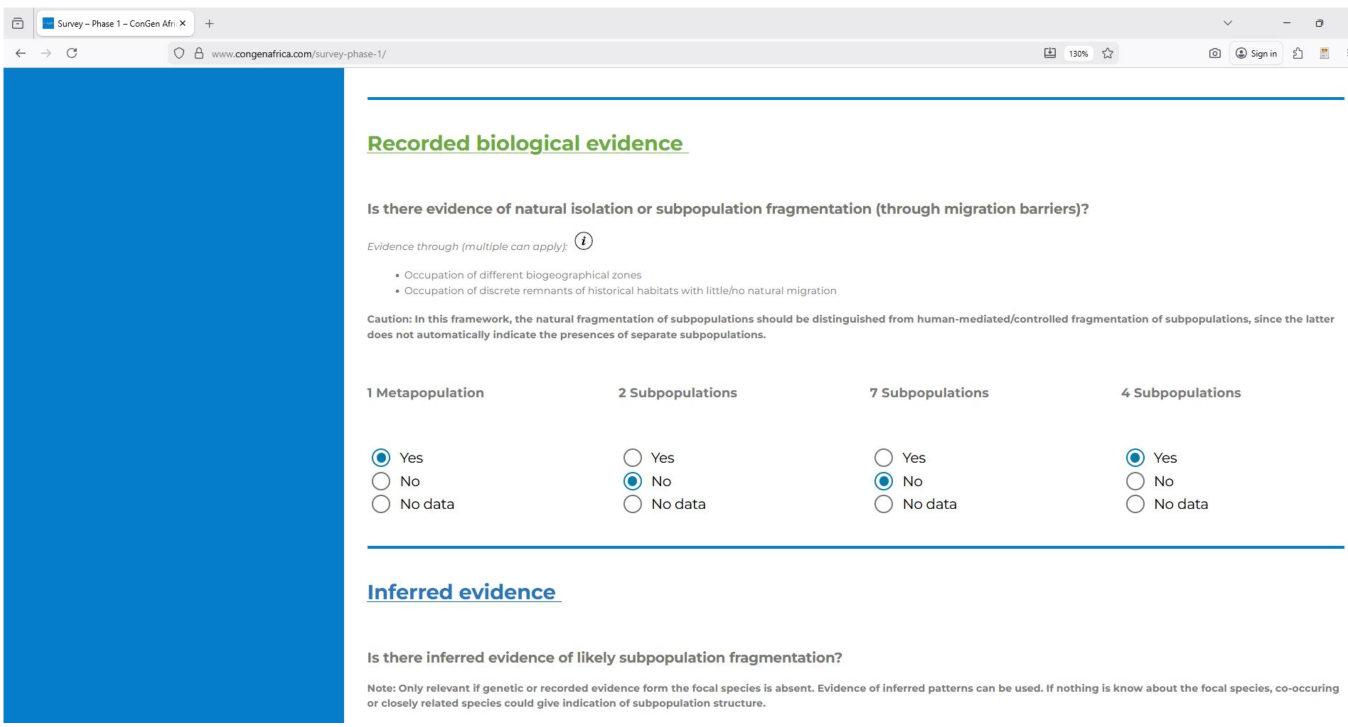


b. Recorded biological evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation. For this scenario scoring is reserved to keep the questions identical across all scenarios. The existing evidence did not indicate a single metapopulation. Therefore, a score of 0/2 for isolation/migration barriers was assigned.

In **Scenarios (2) and (3)**, despite the presence of genetic structuring, the species occurs naturally across the four main Hawaiian islands (Island of Hawaiʻi, Maui, Oʻahu, and Kauaʻi). The observed genetic patterns did not clearly correspond to identifiable migration barriers. Therefore, both scenarios received a score of 0/2 for isolation/migration barriers.

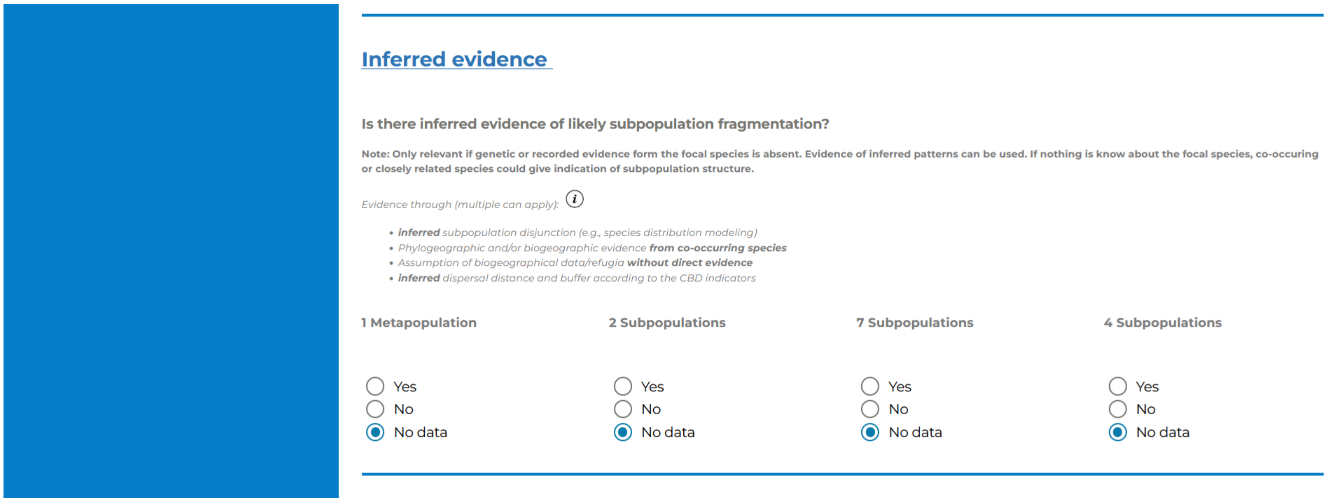
In **Scenario (4)**, the natural occurrence pattern showed distinct occupation of different habitats across islands, which can be interpreted as biogeographical separation and potential migration barriers. This resulted in a score of 2/2 for isolation/migration barriers.



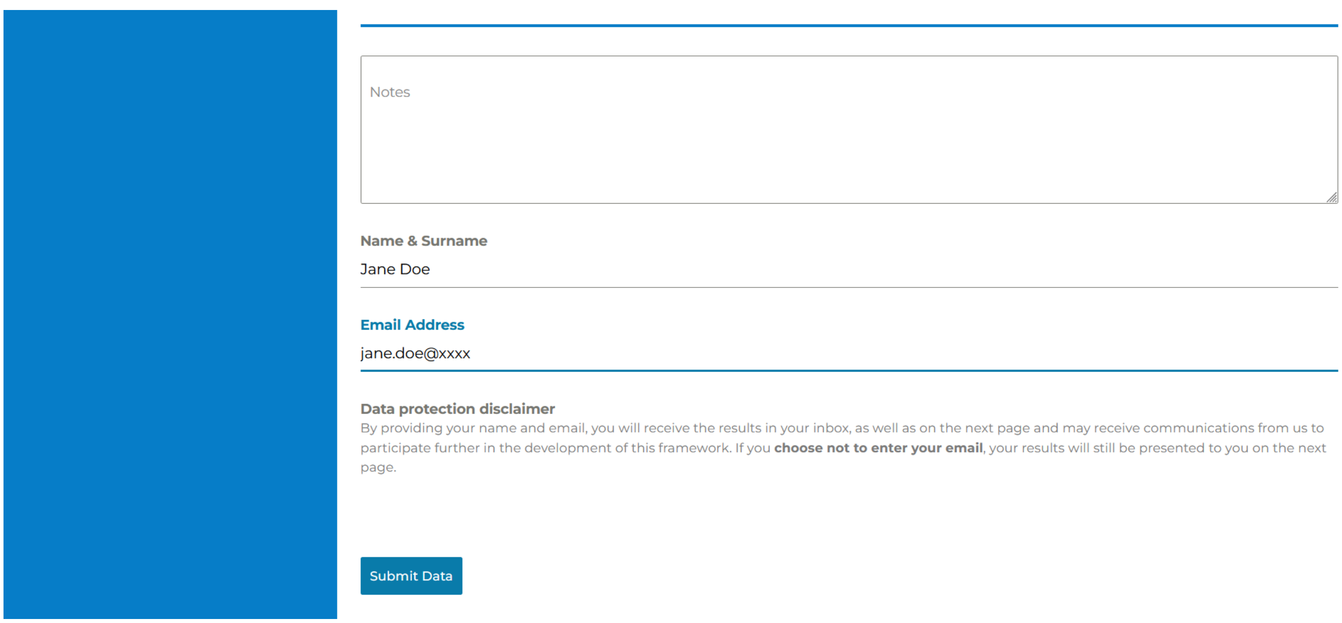
c. Inferred evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation. For this scenario scoring is reserved to keep the questions identical across all scenarios. No inferred evidence was found to support the null hypothesis of a single metapopulation. Therefore, this line of evidence was marked as ‘No data’ and score 0/2 points.

In **Scenarios (2), (3), and (4)**, no inferred evidence was found to support subpopulation fragmentation. As a result, each scenario received a score of 0/2 in this category, and the evidence was marked as ‘No data’.

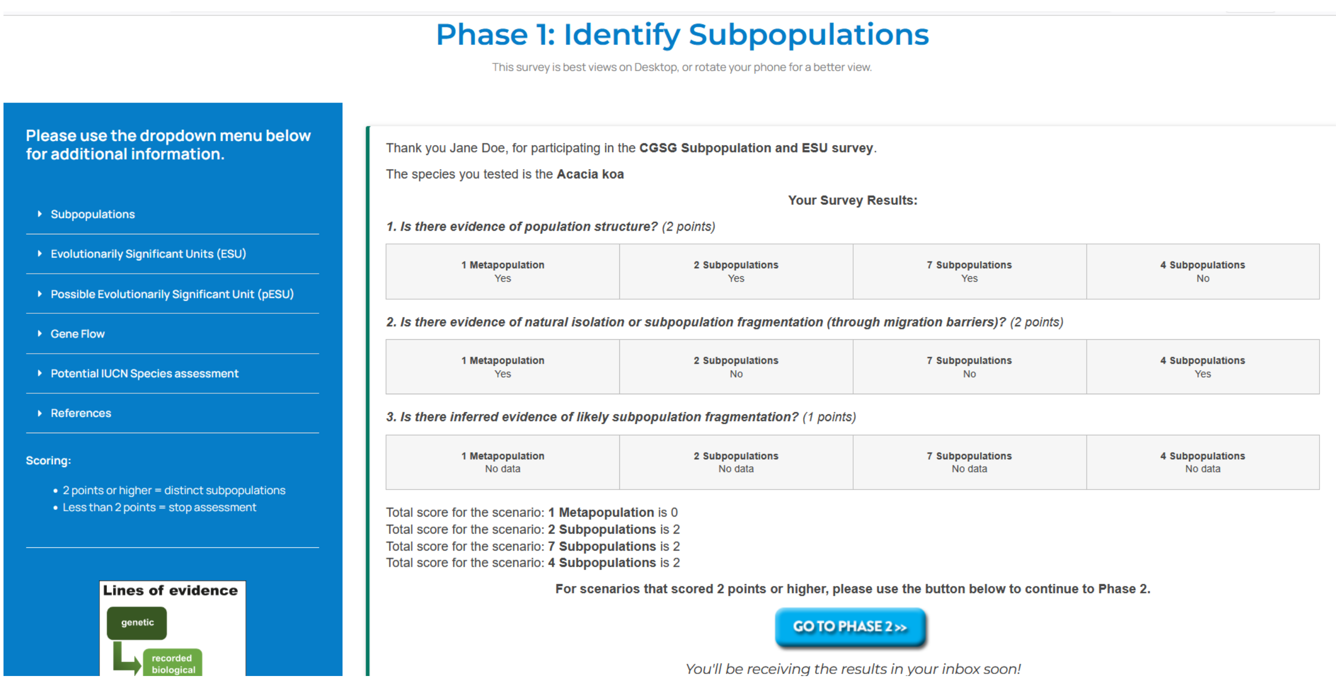


3. Provide name and e-mail address (optional). This will allow the results to be sent to your email address.



4. Click the **“Submit Data” button** to receive the results (mandatory to get results).

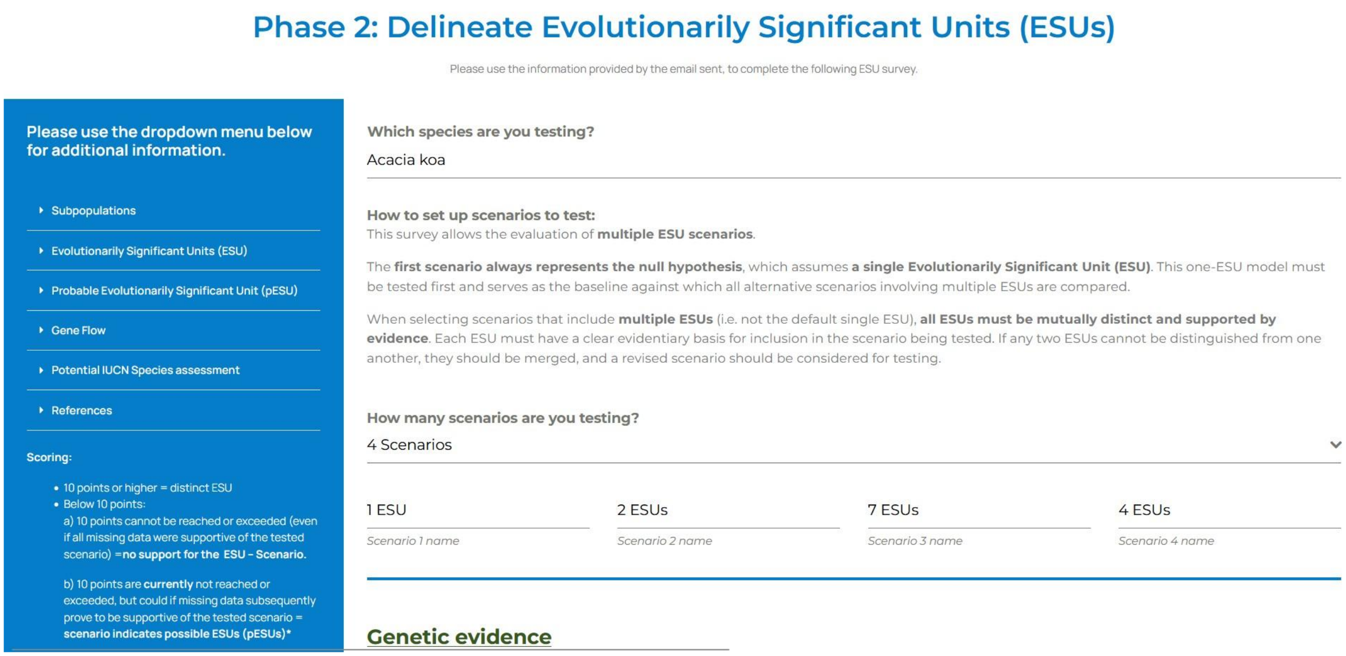
5. Interpreting the results.
   This is the results page. It lists each question and the responses entered by the user as verification. Below the questions is a summary of the scores for each scenario.
   For the Hawaiian koa tree, three of the four candidate scenarios (**Scenarios (2), (3), and (4)**) meet the minimum threshold of two points, and therefore qualify as subpopulation scenarios. Where scores are tied, we recommend selecting the one scenario which is based on the most robust data to maximize the conservation of genetic diversity. In this case, Scenario (3), based on high-resolution genomic markers (11,000 SNPs from >300 samples via genotyping-by-sequencing, Gugger et al., 2018), identified seven distinct subpopulations, while using nine microsatellites from 170 samples (Fredua-Agyeman et al., 2008), identified only two subpopulations. Although microsatellites remain valuable in population genetics, the two-subpopulation scenario (2) is unlikely to represent the true subpopulation structure. Instead, it might better reflect higher-level structuring at the ESU level, along with Scenario (4) (4 islands subpopulations). Overall, the seven-subpopulation scenario (Scenario 3) better represents subpopulation structure.
   Scenarios (2), (3), and (4)) scored equally higher (two points) and therefore can be tested in Phase 2.



6. Click on the **"Phase 2" button** to proceed.

7. As in Phase 1, enter the species and scenario information. This does not get carried over from Phase 1.

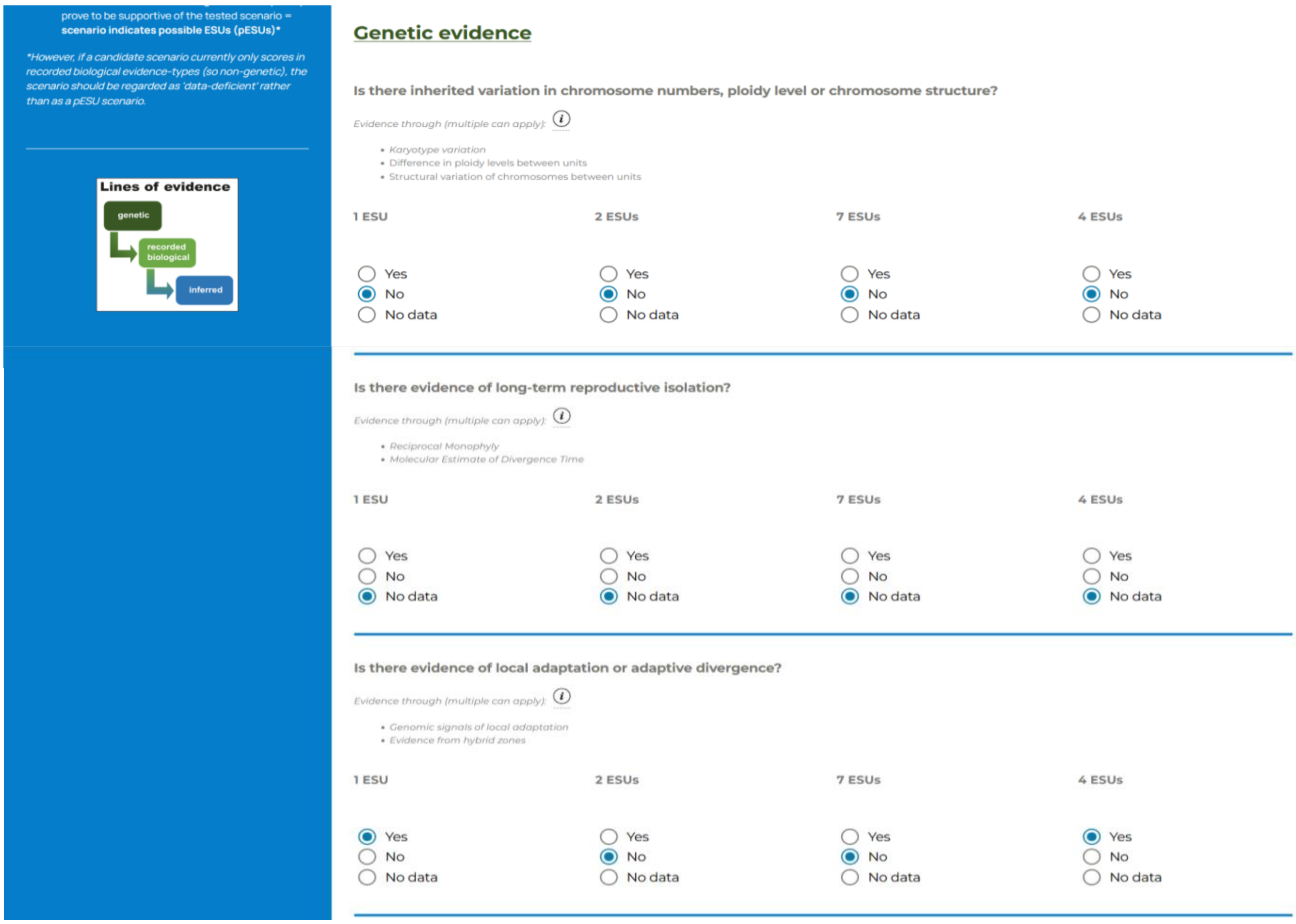
   Based on the results of Phase 1, we tested four candidate scenarios for delineating Evolutionarily Significant Units (ESUs): (1) a single ESU (2) two ESUs (3) seven ESUs and (4) four ESUs.



8. Answer the questions for each line of evidence for each scenario being investigated.
   a. Genetic evidence
      In **Scenario (1)** (null hypothesis), depicting a single ESU, scoring was reserved to maintain consistency across scenarios. A cytology study found tetraploidy (2n=52) across all islands (Shi, 2003) resulting in 6/6 for karyotype variation. However, no evidence for evolutionary distinctiveness was found (0/6). Landscape genomics showed divergence correlated with precipitation (Gugger et al., 2018) and island-based seed zones are supported (Dudley et al., 2017), contradicting a single ESU and resulting in 0/6 for adaptive divergence.
      In **Scenarios (2),** based on two ESUs, tetraploidy across islands (Shi, 2003) resulted in 0/6 for karyotype variation. No evidence for evolutionary distinctiveness was found (0/6). Landscape genomics did not support adaptation patterns matching two ESUs (Gugger et al., 2018), resulting in 0/6 for adaptive divergence.
      In **Scenario (3)**, based on seven ESUs, tetraploidy across islands (Shi, 2003) resulted in 0/6 for karyotype variation. Admixture among subpopulations (Gugger et al. 2018) contradicts ESU criteria of restricted gene flow (leading to unique evolutionary trajectories), resulting in

a score of 0/6 for evolutionary distinctiveness. No adaptive divergence matching this scenario was found (0/6) (Gugger et al., 2018).

In **Scenario (4)**, based on four ESUs corresponding to the main Hawaiian islands, tetraploidy across islands (Shi, 2003) resulted in 0/6 for karyotype variation. No evidence for evolutionary distinctiveness was found (0/6). Genetic divergence correlated with precipitation (Gugger et al., 2018), and island-based seed zones are supported (Dudley et al., 2017), resulting in 6/6 for adaptive divergence.



b) Recorded biological evidence

In **Scenario (1)** (null hypothesis), depicting a single ESU, provenance tests revealed inherited phenotypic differences among islands (Baker et al., 2009), yielding 0/4 for inherited variation. No evidence was found for recorded variation or traditional knowledge (0/2 each).

In **Scenarios (2) and (3),** provenance tests revealed variation at the island level rather than matching the proposed ESU structure (Baker et al., 2009), resulting in 0/4 for inherited variation. No evidence was found for recorded variation or traditional knowledge in either scenario (0/2 each).

In **Scenario (4)**, provenance tests showed inherited phenotypic differences among islands (Baker et al., 2009), yielding 4/4 for inherited variation. No evidence was found for recorded characteristic variation or traditional knowledge (0/2 each).

Recorded biological evidence

Is there evidence of inherited characteristic differences?

Evidence through (multiple can apply):

- *Heritable differences in focal traits observed in experiments or confirmed by analytical tests*

1 ESU	2 ESUs	7 ESUs	4 ESUs
◉ Yes	○ Yes	○ Yes	◉ Yes
○ No	◉ No	◉ No	○ No
○ No data	○ No data	○ No data	○ No data

Is there evidence supporting transmitted characteristic differences?

Evidence through (multiple can apply):

- *Cultural or learnt behavioral differences unique or specific between units*
- Acquired or transmitted traits (e.g., migration pattern, methylation differences, body size differences)

1 ESU	2 ESUs	7 ESUs	4 ESUs
○ Yes	○ Yes	○ Yes	○ Yes
○ No	○ No	○ No	○ No
◉ No data	◉ No data	◉ No data	◉ No data

Is there traditional knowledge suggesting distinctiveness between units?

Evidence through (multiple can apply):

- *Recorded differences based on traditional and local knowledge*

1 ESU	2 ESUs	7 ESUs	4 ESUs
○ Yes	○ Yes	○ Yes	○ Yes
○ No	○ No	○ No	○ No
◉ No data	◉ No data	◉ No data	◉ No data

9. Enter your name and e-mail address (optional).

Notes

Name & Surname

Jane Doe

Email Address

jane.doe@xxxx

Data protection disclaimer

By providing your name and email, you will receive the results in your inbox, as well as on the next page and may receive communications from us to participate further in the development of this framework. If you **choose not to enter your email**, your results will still be presented to you on the next page.

Submit Data

10. Click the **“Submit Data” button** to receive the results.

11. Interpreting the results.

As with the subpopulation survey results, each question and the answers provided are presented on the results page. A summary of the total scores for each scenario is presented below that.

Only scenario (4) (four ESUs) scored 10 points out of 16 points evaluated (with available data), providing strong support for four distinct ESUs. This score combines one genetic line of evidence (‘Adaptive divergence’) with one recorded line of evidence (‘Inherited characteristic variation’). Scenario (1) received 6 points out of 16 evaluated and scenario (2) (two ESUs) and scenario (3) (seven ESUs) received zero points due to lack of supporting evidence.

Phase 2: Delineate Evolutionarily Significant Units (ESUs)

Please use the information provided by the email sent, to complete the following ESU survey.

Please use the dropdown menu below for additional information.

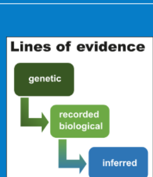
- Subpopulations
- Evolutionarily Significant Units (ESU)
- Probable Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 10 points or higher = distinct ESU
- Below 10 points:
  a) 10 points cannot be reached or exceeded (even if all missing data were supportive of the tested scenario) =**no support for the ESU - Scenario.**

  b) 10 points are **currently** not reached or exceeded, but could if missing data subsequently prove to be supportive of the tested scenario = **scenario indicates possible ESUs (pESUs)***

**However, if a candidate scenario currently only scores in recorded biological evidence-types (so non-genetic), the scenario should be regarded as 'data-deficient' rather than as a pESU scenario.*



Thank you Jane Doe, for participating in the **CGSG Subpopulation and ESU survey**.

The species you tested is the **Acacia koa**

Your Survey Results for Phase 2:

1. Is there inherited variation in chromosome numbers, ploidy level or chromosome structure? *(6 points)*

1 ESU	2 ESUs	7 ESUs	4 ESUs
No	No	No	No

2. Is there evidence of long-term reproductive isolation? *(6 points)*

1 ESU	2 ESUs	7 ESUs	4 ESUs
No data	No data	No data	No data

3. Is there evidence of local adaptation or adaptive divergence? *(6 points)*

1 ESU	2 ESUs	7 ESUs	4 ESUs
Yes	No	No	Yes

4. Is there evidence of inherited characteristic differences? *(4 points)*

1 ESU	2 ESUs	7 ESUs	4 ESUs
Yes	No	No	Yes

5. Is there evidence supporting transmitted characteristic differences? *(2 points)*

1 ESU	2 ESUs	7 ESUs	4 ESUs
No data	No data	No data	No data

6. Is there traditional knowledge suggesting distinctiveness between units? *(2 points)*

1 ESU	2 ESUs	7 ESUs	4 ESUs
No data	No data	No data	No data

Total score for the scenario: **1 ESU** is 6
Total score for the scenario: **2 ESUs** is 0
Total score for the scenario: **7 ESUs** is 0
Total score for the scenario: **4 ESUs** is 10

Scenario scoring:

- Scenarios scoring 10 points or higher = distinct ESUs
- Scenarios scoring below 10 points:
  - **without** potential to reach/exceed 10 points = no support for tested scenario
  - **with** potential to reach/exceed 10 points = possible ESU (pESU) status

You'll be receiving the results in your inbox soon!

You've reached the end of the survey. We trust that the information will provide insight into the management of your species.

References

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